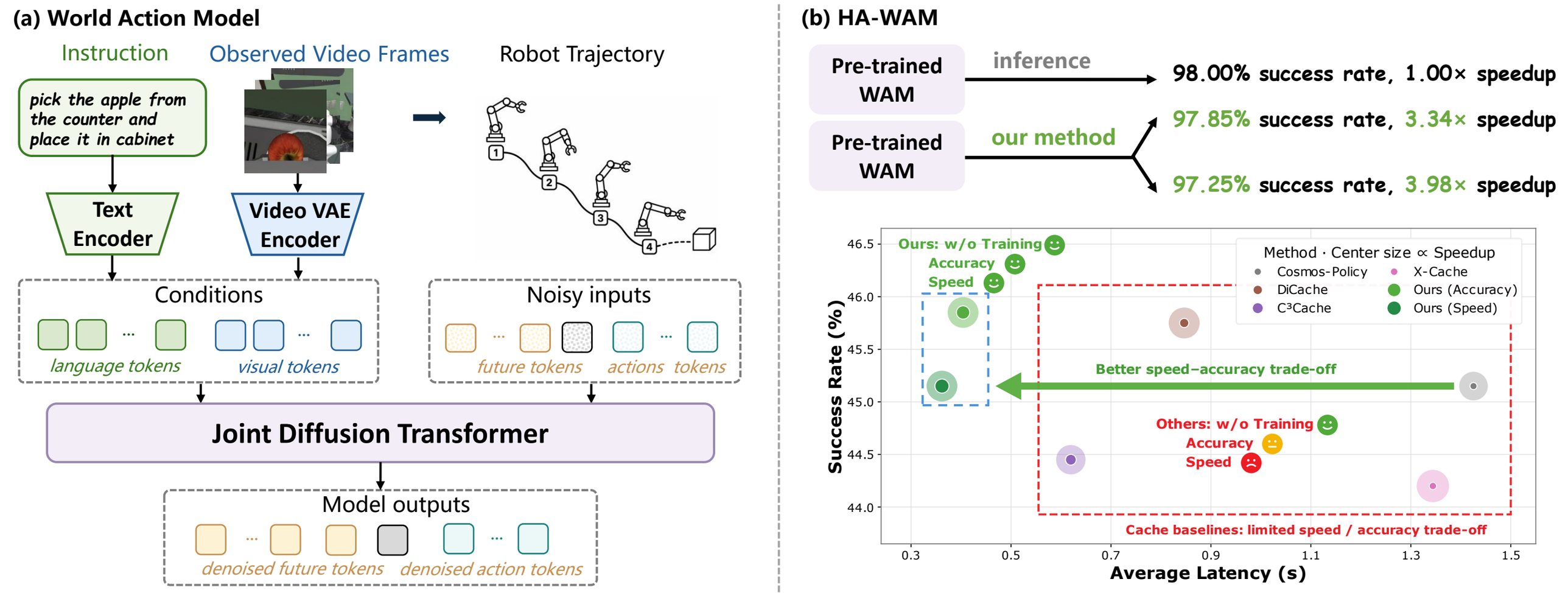

(a) World Action Model
Instruction
Observed Video Frames
Robot Trajectory
pick the apple from the counter and place it in cabinet
Text Encoder
Video VAE Encoder
Conditions
language tokens
visual tokens
Noisy inputs
future tokens
actions tokens
Joint Diffusion Transformer
Model outputs
denoised future tokens
denoised action tokens
(b) HA-WAM
Pre-trained WAM
inference
98.00% success rate, 1.00× speedup
Pre-trained WAM
our method
97.85% success rate, 3.34× speedup
97.25% success rate, 3.98× speedup
Ours: w/o Training
Accuracy
Speed
Method · Center size ∝ Speedup
Cosmos-Policy
DiCache
C³Cache
X-Cache
Ours (Accuracy)
Ours (Speed)
Better speed–accuracy trade-off
Others: w/o Training
Accuracy
Speed
Cache baselines: limited speed / accuracy trade-off
Success Rate (%)
Average Latency (s)
46.5
46.0
45.5
45.0
44.5
44.0
0.3
0.5
0.7
0.9
1.1
1.3
1.5